A paleomagnetic study of the layered mafic intrusion at Sept-Îles, Quebec[1]

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A paleomagnetic study of gabbroic and anorthositic members of the Sept-Îles layered mafic intrusion has isolated a remanence with direction $D = 333°$, $I = -29°$ (remanence A) and a corresponding pole of 141°E, 20°N. The rocks are cut by numerous diabase dykes of unknown age. The remanence carried by the dykes has direction $D = 188°$, $I = -85°$ (remanence B), with a corresponding pole of 116°E, 59°S. Another remanence, statistically identical to B ($D = 186°$, $I = -85°$), is found at dyke contacts and in the local host rock; its associated pole, 115°E, 61°S, is identical to the one derived from the dykes. This overprint is significantly different from remanence A, and is obviously related to dyke emplacement. The location of the pole derived from remanence A is in excellent agreement with many other Cambrian poles from a variety of locations throughout cratonic North America. The rocks at Sept-Îles have been previously dated radiometrically at 540 Ma. Thus, all evidence indicates that remanence A is an original thermochemical remanent magnetization, acquired during initial cooling of the intrusion in the Cambrian and prior to the emplacement of the dykes.

Dans le cadre d'une étude paléomagnétique des phases gabbroïque et anorthositique de l'intrusion stratiforme mafique des Sept-Îles, une aimantation rémanente a été isolée; cette aimantation a une direction $D = 333°$, $I = -29°$ (aimantation A) et un pôle magnétique correspondant de 141°E, 20°N. Cette intrusion est entrecoupée par de nombreux filons de diabase d'âge incertain. L'aimantation rémanente que l'on retrouve dans les filons a une direction de $D = 188°$, $I = -85°$ (aimantation B) qui correspond à un pôle de 116°E, 59°S. Une autre aimantation rémanente qui est statistiquement identique à l'aimantation B a été isolée dans la roche encaissante et ce, au contact des filons de diabase; cette aimantation a un pôle de 115°E, 61°S identique à celui trouvé dans les filons. Cette reaimantation statistiquement différente de l'aimantation A est reliée à l'emplacement des filons. La localisation du pôle dérivé de l'aimantation A concorde avec plusieurs pôles d'âge cambrien obtenus de roches provenant du craton nord-américain. Les roches échantillonnées à Sept-Îles ont été datées par méthode géochronologique à 540 Ma. Toutes les évidences concordent et démontrent que l'aimantation rémanente A est une aimantation initiale d'origine thermo-chimique qui a été acquise lors du refroidissement de l'intrusion de Sept-Îles au Cambrien et ce, avant l'intrusion des filons de diabase.

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Introduction

The Sept-Îles layered mafic intrusion is an attractive target for paleomagnetic study because of its Cambrian (540 Ma) age and its suitability for remanence determination due to high magnetite content. Rocks of this age that are undisturbed by Appalachian tectonics are difficult to find in eastern North America. As a result, the paleomagnetic data base from lower Paleozoic rocks from this part of the craton is relatively scanty. Interpretations of the available poles are hampered by a lack of accurate age determinations, by uncertainty about the integrity of the North American craton, and by the possible complexity of the lower Paleozoic apparent polar wander (APW) path. The Sept-Îles intrusion was studied in order to improve the data base with the intent of eventually establishing a reference APW path for the pre-Appalachian North American craton. This study presents the magnetic history of the gabbroic and anorthositic phases of the intrusion, and some diabase dykes that are possibly related to the same magmatic event.

Geological setting

The Sept-Îles intrusion cuts rocks of the Grenville Structural Province. It is exposed on the north shore of the St. Lawrence River and on the nearby islands. Xenoliths of Grenvillian gneiss are abundant in the gabbro at the contact with the country rock. The intrusion is invaded by numerous dykes of both acidic and basic composition. Acidic dykes and veins are cut by diabase, which represents the last igneous event that affected the area. The locally exposed Ordovician limestone and Grenville gneiss are devoid of diabase intrusions. Thus, field evidence suggests that the diabase dykes followed not long after the main intrusive event (T. Feininger, personal communication, 1986). The dykes and some basic members of the intrusion appear to be comagmatic on geochemical grounds (Higgins and Doig 1986). In the absence of conclusive evidence as to the age of diabase activity, paleomagnetic results from dykes of this study are used primarily for a contact test.

The Sept-Îles intrusion was originally mapped as a Grenvillian anorthosite (Faessler 1942). However, the absence in it of a typical Grenvillian deformation was first noted by Wynne-Edwards (1972), which is fully supported by our field observa-

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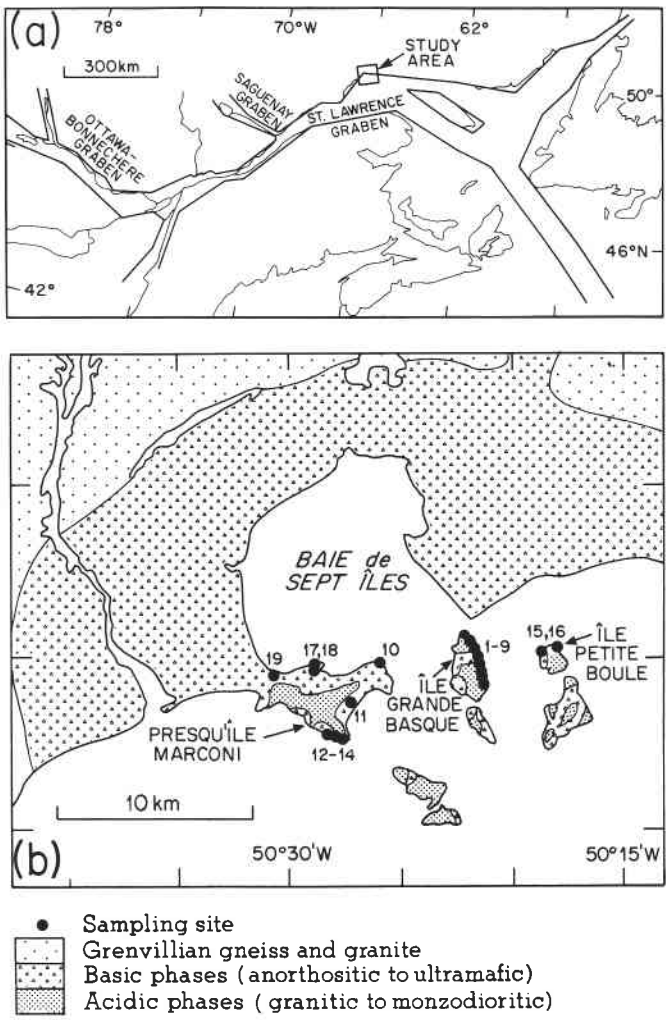


FIG. 1. (*a*) Location of the study area in relation to the St. Lawrence graben system. (*b*) Geology and sampling sites of the Sept-Îles intrusion. (Map adapted from Higgins (1979) and Franconi *et al.* (1975).)

tions. Weak fabrics can be detected in the immediate vicinity of some faults, but these die out within metres of the disturbance. The intrusion differs greatly from the anorthosites found inland to the north of the St. Lawrence River. Recent mapping by Feininger demonstrates that the body is a layered mafic intrusion, consisting of many petrological phases, only one of which is a true anorthosite. The body is roughly circular in shape and coincides with a prominent gravity high (Goodacre 1976). The shape and spacing of the gravity contours suggest that the body is funnel shaped and deeply rooted.

The results of a reconnaissance survey by Fahrig (cited in Goodacre 1976) indicated that remanent magnetizations characteristic of the intrusion bear more similarity to Cambro-Ordovician than to Grenvillian pole positions. The age of the intrusion was finally established as Cambrian by the Rb–Sr method (Higgins 1979; Higgins and Doig 1981). Six isochrons on both acidic and basic members of the intrusion yielded an average age of 540 ± 22 Ma (decay constant 1.42×10^{-11} a^{-1}). Similar initial ratios of $^{87}Sr/^{86}Sr$ suggested that all rocks were derived from the same parent magma. In this time period, intrusion of alkalic complexes was widespread in eastern Canada. Kumarapeli (1976) explained their emplacement as a consequence of active rifting along the St. Lawrence graben. As the Sept-Îles intrusion is also situated on this proposed zone of crustal weakness (Fig. 1*a*), its origin has been attributed to the same cause (Doig 1970; Kumarapeli and Saull 1966).

No main fault boundary is evident in the immediate area; however, a change of lithology is observed between Presqu'île Marconi and the mainland (Fig. 1*b*). This is interpreted as the slumping of the intrusion in the centre after the roof had already solidified and while the lower zone was still plastic and in a state of tension. Presqu'île Marconi and the islands apparently subsided as a single unit with respect to the coast (Higgins and Doig 1981). The result is deeper exposure around the edge of the intrusion, manifested by basic members such as layered and massive gabbro, troctolite, pyroxenite, and anorthosite. With the exception of small veins, acidic rocks are totally absent from the mainland. In contrast, on Presqu'île Marconi and the islands, the only basic phases are massive gabbro and anorthosite. These are overlain by a thick sequence of monzonite, syenite, and granite, which form high topography. A detailed distribution of rock types is not available to date, and for the purposes of this paper, the lithologies are grouped into two broad categories: acidic and basic (Fig. 1*b*).

Igneous layering is subhorizontal everywhere within the subsided centre of the intrusion. For this reason it was not necessary to apply tilt corrections to the paleomagnetic results. On the island of Grande Basque the dip of the contact between anorthosite and monzonite varies locally from 0 to 10°E. It is a zone, 2 m thick, containing pillows of anorthosite in the overlying syenite. A gentle southerly dip of this contact was reported by Higgins (1979). Slight tilting would be consistent with downdrop towards the southeast along normal fault planes. However, dips could also be attributed to original irregularities of the igneous layering. Examples of slumping and turbulence at phase contacts are abundant throughout the Sept-Îles intrusion. No postemplacement tilting appears to be present. Therefore, the sampling sites have been considered *in situ* with respect to horizontal for paleomagnetic purposes.

Methodology

Paleomagnetic sampling of the Sept-Îles intrusion was concentrated on the islands and Presqu'île Marconi, where outcrop is abundant, and subhorizontal igneous layering provides good tilt control. A total of 118 cores was taken from 19 localities on Presqu'île Marconi and the islands (Fig. 1*b*, Table 1). Of those, sites 2, 6, 12, and 18 were drilled in diabase dykes. Two to four specimens were obtained from each core. Thermal and alternating field (AF) demagnetization were performed on the collection. The measurements were made on a Schonstedt digital spinner magnetometer (DSM), and the ensuing calculations and plots carried out on an Interdata Model 70 computer (Lapointe and Carrière 1978). For a comprehensive description of the paleomagnetic methods used in this laboratory, the reader is referred to the discussion of methodology in Roy and Anderson (1981).

Natural remanent magnetization (NRM)

Anorthositic complexes are typically enriched in primary magnetite and other iron-bearing minerals (Philpotts 1966), and the Sept-Îles intrusion is no exception. Polished thin section and microprobe studies by Higgins (1979) have detected the presence of fine-grained magnetite throughout the basic phases. Specks of the mineral are present in the unzoned cores

TABLE 1. Results of demagnetization

Site	Distance to dyke contact	Remanence A							Remanence B					
		D (deg)	I (deg)	N	k	α_{95} (deg)	afs (mT)	T_{UB} (°C)	D (deg)	I (deg)	N	k	α_{95} (deg)	T_{UB} (°C)
1	25 cm to site 2	320	−34	13	61	5	120	560	—	—	—	—	—	—
2		—	—	—	—	—	—	—	008	80	8	32	10	540
3		336	−40	9	61	7	75−150	570	—	—	—	—	—	—
4		331	−41	3	16	31	—	570	256	−79	4	22	20	500
5	10 cm to site 6	—	—	—	—	—	—	—	174	−88	2	—	—	530
6		—	—	—	—	—	—	—	262	89	8	18	13	550
7		—	—	—	—	—	—	—	347	65	6	126	6	500
8		334	−21	5	21	17	75−150	570	031	83	2	—	—	500
9		—	—	—	—	—	—	—	038	71	6	10	22	525
10		342	−24	3	57	16	75	540	023	−78	4	30	17	550
11		326	−39	6	100	8	75−120	550	254	−88	2	—	—	540
12		—	—	—	—	—	—	—	028	−83	8	32	10	550
13	5 cm to site 12	323	−30	4	28	18	50−110	—	263	−88	2	—	—	560
14		330	−10	5	173	6	—	570	—	—	—	—	—	—
15		344	−38	2	—	—	40	580	025	85	6	23	14	550
16		—	—	—	—	—	—	—	349	76	4	94	10	525
17		343	−10	1	—	—	—	550	021	79	4	27	18	550
18		—	—	—	—	—	—	—	019	77	6	25	14	555
19		—	—	—	—	—	—	—	073	−73	4	80	10	530

NOTES: D and I = declination and inclination with respect to horizontal; N = number of samples; k = precision parameter; α_{95} = radius of circle of confidence; *afs* = alternating field stability; T_{UB} = unblocking temperature.

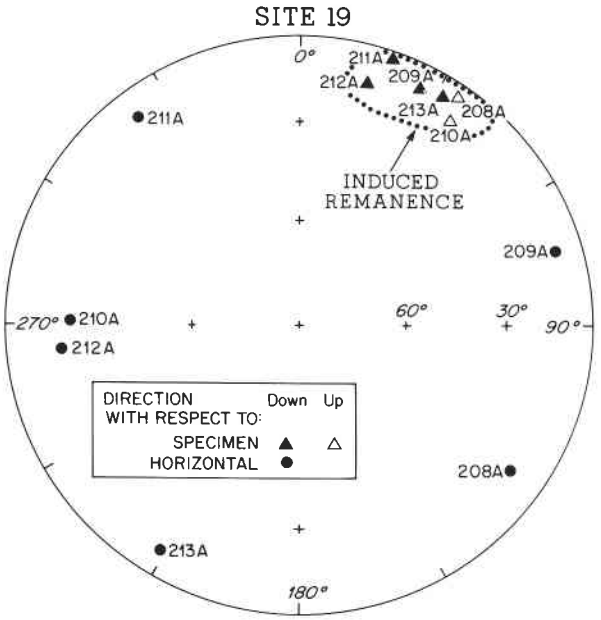


FIG. 2. Example of induced remanences. Declination and inclination with respect to randomly oriented specimens are grouped in the NE quadrant. Correcting all remanence vectors to the horizontal of the sampling locality results in a random distribution.

of plagioclase crystals, but not in the rims, suggesting a primary igneous origin.

High intensities of NRM are relatively homogeneous throughout all studied sites. The intensities range between 1.0 and 0.1 A/m and are typical of rocks containing abundant magnetite. A well-defined NRM grouping indicates that the magnetization of the rocks is not easily disrupted, and, therefore, stable demagnetization results can be expected. Polarities of NRM's are mixed in the majority of sites. However, sites characterized exclusively by negative inclinations tend to reveal, upon demagnetization, a characteristic remanence, denoted as A. Positive sites and sites of mixed polarities are associated with a second characteristic remanence, denoted as B.

Results of thermal demagnetization

The major part of the collection was demagnetized thermally up to 700°C in incremental steps of 10° above 500°C. Between 550° and 580°C the interval was often reduced to 5°. Although many specimens show strong remanences, it can be shown that all results above approximately 580°C can be attributed to magnetizations induced between treatment and measurement procedures. This high-temperature component appears throughout the collection soon after the previous remanence is unblocked, and remains unchanged, often to as high as 700°C (see example Fig. 4*c*). After correction for core orientation no statistically significant site mean directions were found. With respect to the specimen, that is, prior to any correction, the direction consistently falls in the northeast quadrant at a shallow inclination, for all sites and samples. A comparison of within-site distribution of the declination and inclination with respect to specimen versus the declination and inclination with respect to the present horizontal was performed for each site. A typical result is shown in Fig. 2 for site 19.[2] The drill holes were randomly oriented, and, as shown after correction for core orientation, the observed remanence directions become random: it is readily apparent that this component cannot have any geological significance. The measuring process is the most likely means of inducing this magnetization with respect to specimen. Samples must be

[2]With the exception of Fig. 2, all remanence vectors are plotted with respect to the present horizontal and true north.

Table 2. Example of contact test

Site	Sample	Distance from contact (m)	*D*, *I* (deg), remanence A	*D*, *I* (deg), remanence B
12	136A			233, 72
(Dyke)	136B			176, 77
	137B			146, 84
	138A			303, 84
	138B			182, 59
	139A			206, 75
	140A			005, 80
	141A			326, 79
13	142A	0.05		Unstable
(Gabbroic	142B	0.05	Unstable	
anorthosite)	143A	0.68		255, 78
	143B	0.68	Unstable	
	144A	3.00		077, 74
	144B	3.00	Unstable	
	144C	3.00	315, −11	
	145B	12.00	318, −40	
	146A	30.00	322, −31	
	147B	50.00	337, −36	

NOTES: Remanences were calculated giving unit weight to treatment steps ($N = 8$ on average). Only those remanences with $\alpha_{95} < 12°$ were accepted. For $\alpha_{95} \geq 12°$, the vector was considered unstable. A blank in the column indicates that the remanence was not found.

removed and replaced in the instrument in a set sequence of orientations. Brief exposure to the Earth's field is inevitable, and because it is identical for all samples, agreement of the induced magnetization is observed with respect to specimen. Any ancient remanence that may be present above 580°C in this anorthosite is either too weak to be detected by the sensitivity of the instrument (5×10^{-4} A/m) or is totally obscured by the much stronger induced component (on the order of 5×10^{-2} A/m).

Below 580°C two remanences, A and B, were isolated by thermal demagnetization. Vector subtraction was performed in all cases where a high-temperature induced magnetization was detected. The results show no significant distortion of the true remanence by the laboratory-acquired magnetization. Remanence A is located in the northwest quadrant and has an intermediate negative inclination (Fig. 3*a*; Tables 1−3). This remanence typically unblocks in the range between 550 and 580°C, with a corresponding abrupt drop of intensity in the decay curve. Such a pattern, along with the high intensity observed below the unblocking temperature (in the order of 9×10^{-1} A/m), is characteristic of magnetite. A second, steeply inclined remanence (Figs. 4*a* and 4*c*; Tables 1−3), superficially resembles a viscous component induced by the present magnetic field. (According to the magnetic repeat station observatory of the Earth Physics Branch, Energy, Mines and Resources Canada, the magnetic field at Sept-Îles was $D = 335°$, $I = 75°$, in 1983). However, remanence B cannot be readily explained as a recent overprint, because it occurs equally in both positive and negative polarities (Figs. 4*b* and 4*c*). A direction reversal upon demagnetization was often observed in the same sample, typically in the 500−525°C range (Fig. 4*a*). In Table 1 remanence B is expressed in the polarity that prevails in the given site. Exclusively one polarity was encountered only at sites 7, 9, 10, and 17.

A striking difference between the two remanences, apart from orientation, is the intensity decay spectrum under thermal treatment. In contrast to the typically discrete decay of remanence A (Fig. 3*a*), the decay of remanence B (Fig. 4*a*) is distributed over the entire range between 100 and 550°C. This observation is supportive of the overprinting nature of remanence B.

The two remanences, A and B, occur with equal frequency throughout the intrusion. In some sites, only one is present; in others, both occur in different samples, often from the same core. The two remanences were never separated in the same sample under thermal treatment. Examination of Table 1 reveals that, although the unblocking temperature (T_{UB}) of remanence B is, on average, slightly lower than that of remanence A, the probability of isolating remanence A in a narrow band ca. 550°C is highly unlikely. Nevertheless, the demagnetization characteristics can be used as a discriminant of relative acquisition age. Therefore, the presence of dual polarity, the more distributed intensity spectra of remanence B, and the slight difference in T_{UB} of the two remanences suggest that remanence B was acquired after remanence A.

Results of alternating field (AF) demagnetization

The direction of remanence A was successfully defined by AF demagnetization (Figs. 3*b* and 3*c*) and agrees fully with the thermal results. However, the intensity decayed gradually under AF treatment, in contrast to the abrupt intensity drop of remanence A under thermal treatment. Typically, part of remanence A resisted alternating fields of 75 mT or higher, and no other remanences were uncovered by treatments up to 270 mT. In view of the thermal results, the magnetic carrier is inferred to be magnetite. This does not contradict the observed high alternating field stability, because if the mineral were sufficiently fine grained or elongate, such stability would be possible (Evans *et al.* 1968). Therefore, the magnetite dust that pervades the plagioclase crystals of the basic phases appears to be the principal carrier of remanence A.

With very few exceptions, remanence B was not detected at all by AF treatment. Some samples from the dyke at site 2 revealed this remanence, identical to the one isolated by

TABLE 3. Average remanence vectors and poles

(*a*) Unit weight given to sites*

Fisher statistics†								Pole position	
N	*R*	*k*	α_{95} (deg)	δ_p (deg)	δ_m (deg)	*D* (deg)	*I* (deg)	Long.	Lat.
				Remanence A (gabbro and anorthosite)					
10	9.732	34	8	5	9	333	−29	141° E	20° N
				Remanence B (gabbro and anorthosite)					
12	11.758	45	7	13	13	186	−85	115° E	61° S
				Remanence B (dykes)					
4	3.963	81	10	20	20	188	−86	116° E	59° S

(*b*) Unit weight given to samples

Fisher statistics								Pole position	
N	*R*	*k*	α_{95} (deg)	δ_p (deg)	δ_m (deg)	*D* (deg)	*I* (deg)	Long.	Lat.
				Remanence A (gabbro and anorthosite)					
51	48.977	25	4	3	5	329	−31	145° E	18° N
				Remanence B (gabbro and anorthosite)					
46	43.560	18	5	10	10	003	−83	115° E	64° S
				Remanence B (dykes)					
30	28.718	23	6	11	11	181	−87	114° E	57° S

*Results from (*a*) used in abstract and in Fig. 5.

†*N* = number of sites or samples; *R* = resultant vector; δ_p and δ_m = semiaxes of the oval of confidence at probability level of 0.05. Remaining parameters are defined in Table 1.

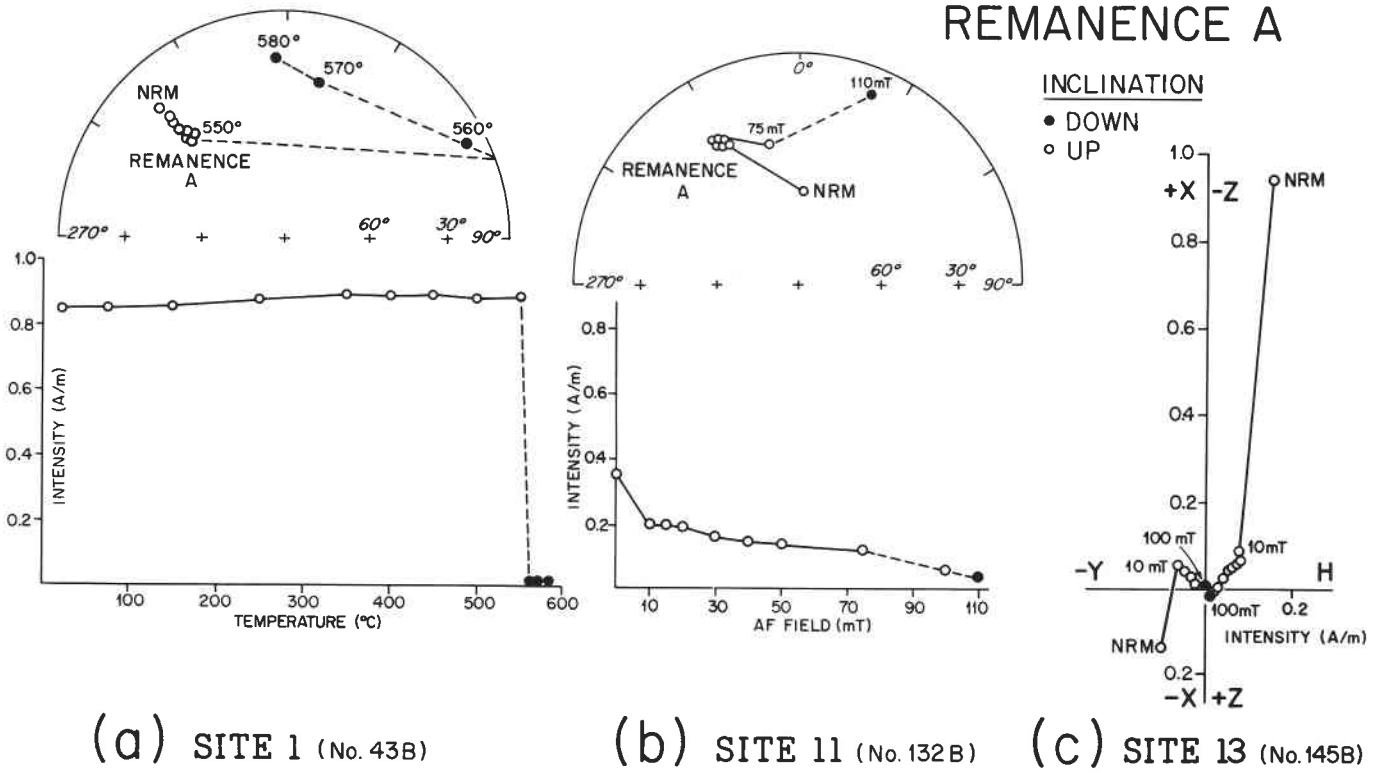


FIG. 3. (*a*) Characteristic thermal demagnetization of remanence A, showing high directional stability to unblocking temperature above 550°C and a corresponding abrupt drop in intensity. (*b*) Characteristic alternating field demagnetization of remanence A, showing high directional stability to 75 mT and gradual intensity decay. (*c*) Remanence A in vector diagrams shows directional stability between 10 and 100 mT. The steep negative southwesterly vector removed between NRM and 10 mT may represent a trace of remanence B.

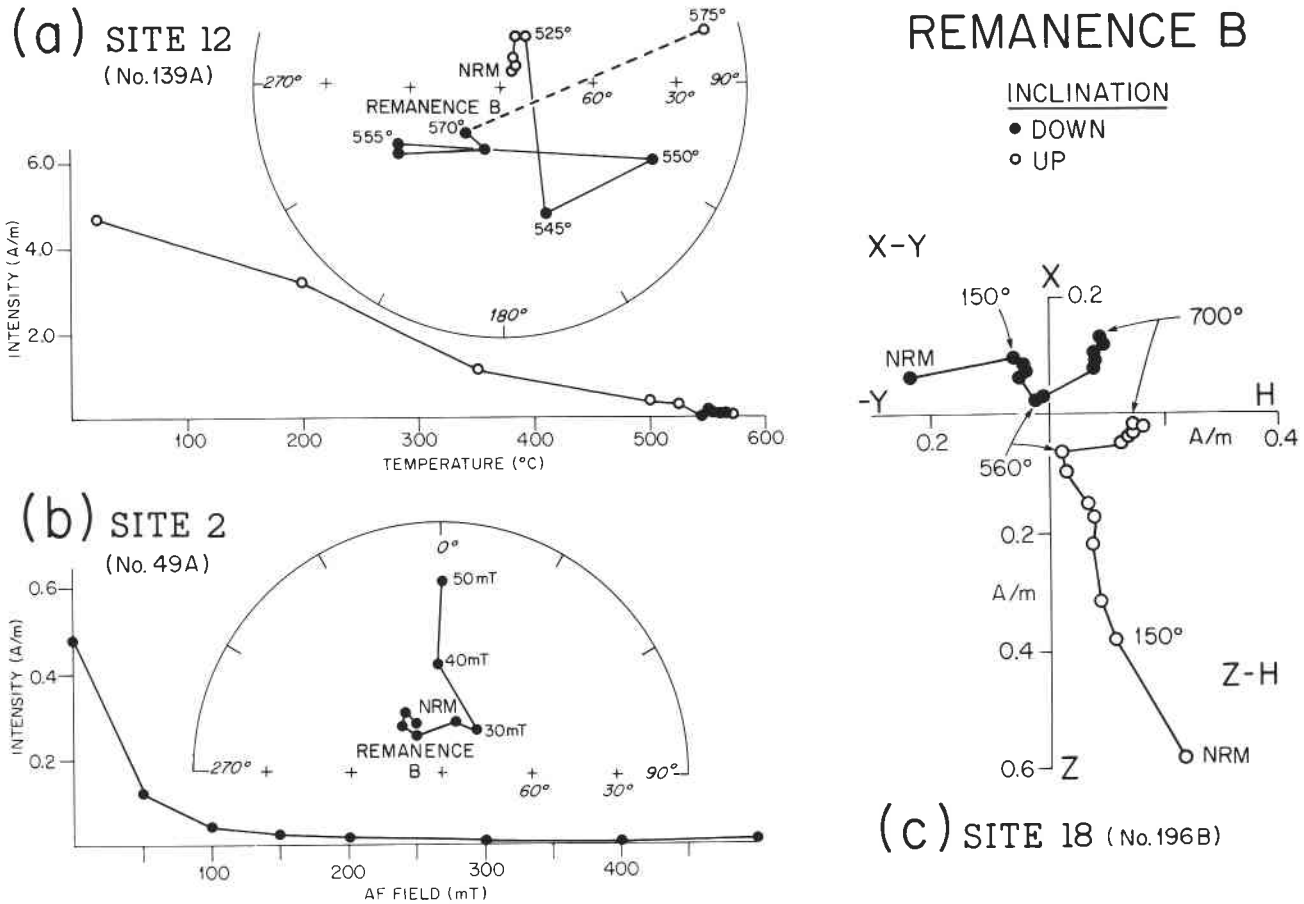


FIG. 4. (*a*) Characteristic thermal demagnetization of remanence B, showing, in this case, a reversal of polarity above 525°C and gradual intensity decay. (*b*) A rare detection of remanence B by the alternating field method, showing a rapid drop in intensity and directional instability above 30 mT. (*c*) In vector diagrams remanence B exhibits a distributed unblocking temperature spectrum (150–560°C). Above 560°C an induced remanence predominates, and the intensity remains approximately constant up to 700°C.

thermal treatment from the same cores (Fig. 4*b*). Typically, the vectors showed signs of instability above 10 mT and by 20 mT the remanence was randomized. Outside these cases, remanence B vanished well below 10 mT, and was recognizable only in the NRM direction. An example of this can be seen in Fig. 3*c*, where both the intensity and direction of the remanence vector change abruptly at the onset of treatment. If this NRM is due to the B overprint, the figure represents a rare case where A and B were separated in the same sample.

Contact test

By contrast with the two distinct remanences recorded in the Sept-Îles intrusion, only one, remanence B, was found in the four dykes sampled at sites 2, 6, 12, and 18. Examination of the sites adjacent to dykes demonstrates the association of remanence B with the dyke intrusion in all four cases. Sites are numbered sequentially as they were drilled along coastlines (Fig. 1). The minimum distance to an adjacent dyke is specified in Table 1, where contacts were sampled. The distance between sites is 50 m or greater in all other cases. The contact test shows the older remanence in the intrusion (A) giving way to the remanence characteristic of the dykes (B) as the dykes were approached.

Sites 12 and 13 from Presqu'île Marconi are the best example of this overprinting process. Detailed results given in Table 2 show the transition occurring in core 144 at 3 m from the geological contact. The magnetic contact apparently runs through this core. It is not surprising that the contact is so sharp, because the magnetic properties of remanences A and B virtually exclude their coexistence.

Sites 1 to 9 were collected along the shore of the island of Grande Basque. Site 3, for example, carries only remanence A. Further along the coast, at site 4, remanences A and B are both present. At site 5, within a metre from a dyke (site 6) no vestige of remanence A remains. Sites 1 and 2 are another dyke and contact pair. Here the transition from remanence A to B takes place abruptly. The core closest to the dyke in site 1 was only 0.25 m from the contact; however, the dyke is also much narrower (0.5 m wide) than that at site 6 (2–3 m). Sites 17 and 18, however, are separated by 50 m, and yet the overprint has affected the entire area. Site 18 includes both a dyke and a 10 m wide contact; the remanences of the dyke and host rock are indistinguishable.

On Presqu'île Marconi the density of intrusion of dykes is much greater than on Grande Basque. Consequently, the regional thermal gradient was probably enhanced, as was the extent of overprinting. On Presqu'île Marconi, only one site (14) carries no effect of remanence B, whereas on Grande Basque, A-type sites are common. Thus, variations in width and abundance of dykes may account for the observed variability in the width of remagnetized contacts.

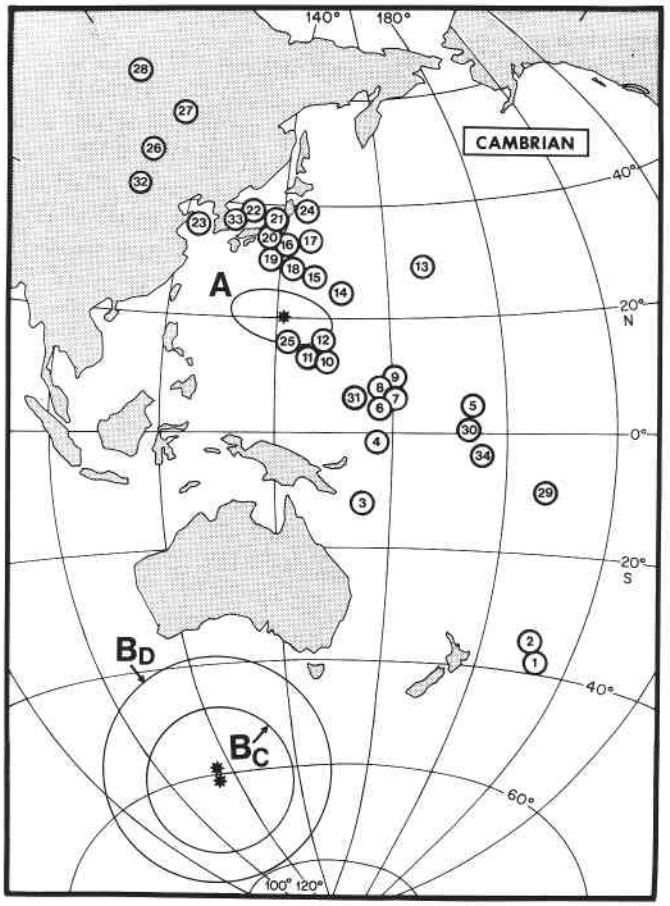


FIG. 5. Cambrian paleomagnetic results for North America. Poles 1–28 as listed in Dankers and Lapointe (1981), poles 29–31 are from Buckingham dyke (Dankers and Lapointe 1981), poles 32 and 33 from Nolichucky Formation (Gillett 1982), and pole 34 from Dunn and Elmore (1985). Poles A, B_C, and B_D of this study are shown with ellipses of confidence.

Conclusions

The pole calculated from remanence A (Table 3) is in close agreement with a concentration of Cambrian poles from widely separated localities in cratonic North America that falls between latitudes 10 and 40°N (Fig. 5). Most of the localities from which these poles are derived are far removed from the effects of Appalachian deformation (Dankers and Lapointe 1981). Therefore, the clustering cannot be attributed to a single tectonometamorphic event in the Paleozoic. A 540 Ma age on the Sept-Îles intrusion by Higgins and Doig (1981) allows the logical connection to be made between the time of acquisition of remanence A and the setting of the Rb–Sr clock. The origin of remanence B can be attributed to the resetting of magnetization at the time of the dyke event, which postdated the emplacement of the intrusion. At some sites specimens with remanence A are found interspersed with B magnetizations. The variable distribution of the B overprint can be explained as a variation in the thermal gradient associated with the dyke intrusion. If the acquisition sequence of the two remanences were reversed, one would need to appeal to localized chemical alterations, for which there is no evidence, to explain the presence of remanence A. Thus, the combination of petrological, radiometric, and paleomagnetic data indicate that remanence A in the Sept-Îles intrusion is a thermochemical remanent magnetization (TCRM),[3] blocked ca. 540 Ma, and the corresponding pole is a useful indicator of the position of the North American craton in the Cambrian.

The timing of remanence B must remain in question until an age is obtained on the dykes. Influxes of basic magma, associated with rifting episodes along the St. Lawrence graben, could have taken place at various times since the Cambrian. The possibility of a viscous overprint dut to the recent field is discounted by the positive contact test and the bipolarity of the remanence, which occurs both in dykes and their remagnetized contacts. Figure 5 demonstrates the close agreement of the two poles B_D and B_C, obtained from dykes and contact respectively. It may be noteworthy that remanence B falls among other high-latitude poles obtained by various workers on some lower Paleozoic rocks, as summarized in Roy *et al.* (1983). The significance of this high-latitude scatter of poles from the lower Paleozoic of North America is not clear owing to a scarcity of data, and remains a subject of discussion (Roy 1982). If complex apparent polar wander is present in this period, much more data will be required for its clarification.

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